

Engineering Mathematics Study Notes: Cauchy's Root Test

1. IDENTIFY

The mathematical concept presented on the board is **Cauchy's Root Test** (frequently called the n -th Root Test). It is an elegant convergence test for infinite series of positive terms. It is uniquely engineered to handle mathematical terms wrapped inside global exponents like n , n^2 , n^3 , or n^4 .

2. THE FORMULA

Let $\sum u_n$ be an infinite series of positive terms where individual terms contain variable powers. We evaluate the limit value l by taking the n -th root of the general term:

$$\lim_{n \rightarrow \infty} (u_n)^{\frac{1}{n}} = l$$

The structural behavior of the infinite series is determined by comparing l directly against 1:

1. The series $\sum u_n$ is **convergent** if:

$$l < 1$$

2. The series $\sum u_n$ is **divergent** if:

$$l > 1$$

3. The test **fails** to provide a conclusion if:

$$l = 1$$

3. THE ALGORITHM

This structural blueprint highlights how to execute Cauchy's Root Test mechanically.

Step 1: Detect the Visual Trigger

Look for general terms that are entirely enclosed within an outer power of n or its powers (n^2 , n^3 , etc.).

Step 2: Extract the General Term

Identify and write out the exact algebraic expression for the n -th term, designated as u_n .

Step 3: Apply the Radical Multiplier

Raise the entire expression u_n to the power of $\frac{1}{n}$.

$$(u_n)^{\frac{1}{n}}$$

Step 4: Neutralize the Outer Exponent

Apply exponent laws to multiply the internal power by $\frac{1}{n}$, canceling out the dominant n variable.

Step 5: Compute the Limit

Evaluate the limit of the remaining structural expression as $n \rightarrow \infty$.

$$l = \lim_{n \rightarrow \infty} (u_n)^{\frac{1}{n}}$$

Step 6: Execute the Decision Rule

Check if the output value l is strictly less than 1 (convergent) or greater than 1 (divergent).

4. THE MECHANICS

Core Operations Explained

- **Power of a Power Rule:** When an exponential expression is raised to another power, the exponents multiply together directly.

$$(a^m)^n = a^{m \times n}$$

- **Exponent Cancellation:** Raising an n -th power to $\frac{1}{n}$ completely strips away the outer exponent.

$$(x^n)^{\frac{1}{n}} = x^{n \times \frac{1}{n}} = x^1 = x$$

- **Higher-Order Exponent Reduction:** If a term is raised to n^2 , multiplying by $\frac{1}{n}$ reduces the power down to a single linear n .

$$(x^{n^2})^{\frac{1}{n}} = x^{n^2 \times \frac{1}{n}} = x^n$$

- **Infinity Fractions:** When evaluating the limit as $n \rightarrow \infty$, terms structured as $\frac{\text{constant}}{n}$ dissolve systematically to 0.

Full Mechanics Walkthrough: Problem 1

Consider the series:

$$\sum_{n=1}^{\infty} \frac{2^n}{3^n}$$

Extract the general term u_n and pull out the shared exponent:

$$u_n = \frac{2^n}{3^n} = \left(\frac{2}{3}\right)^n$$

Apply the n -th root operator to the term expression:

$$\lim_{n \rightarrow \infty} (u_n)^{\frac{1}{n}} = \lim_{n \rightarrow \infty} \left[\left(\frac{2}{3}\right)^n \right]^{\frac{1}{n}}$$

Multiply the exponents together to simplify the expression layout:

$$\lim_{n \rightarrow \infty} \left(\frac{2}{3}\right)^{n \times \frac{1}{n}}$$

Cancel the inverse matching variable terms in the exponent:

$$\lim_{n \rightarrow \infty} \left(\frac{2}{3}\right)^1$$

Evaluate the constant limit directly:

$$l = \frac{2}{3}$$

Compare the final evaluated value against the threshold rule:

$$\frac{2}{3} < 1$$

Since the limit value satisfies $l < 1$, the series is **convergent**.

Full Mechanics Walkthrough: Problem 2

Consider the series:

$$\left(\frac{1}{3}\right)^1 + \left(\frac{2}{5}\right)^2 + \left(\frac{3}{7}\right)^3 + \dots$$

Determine the general mathematical pattern for the n -th term u_n :

- Numerator pattern (1, 2, 3 ...) maps to: n
- Denominator pattern (3, 5, 7 ...) maps to an arithmetic progression: $2n + 1$
- Global power pattern (1, 2, 3 ...) maps to: n

$$u_n = \left(\frac{n}{2n+1}\right)^n$$

Apply the root test operator to the constructed expression:

$$\lim_{n \rightarrow \infty} (u_n)^{\frac{1}{n}} = \lim_{n \rightarrow \infty} \left[\left(\frac{n}{2n+1}\right)^n \right]^{\frac{1}{n}}$$

Execute exponent multiplication to remove the outer bracket power:

$$\lim_{n \rightarrow \infty} \left(\frac{n}{2n+1}\right)^{n \times \frac{1}{n}}$$

Simplify the expression down to its core fraction component:

$$\lim_{n \rightarrow \infty} \frac{n}{2n+1}$$

Factor out the dominant variable n from the internal terms of the denominator:

$$\lim_{n \rightarrow \infty} \frac{n}{n \left(2 + \frac{1}{n}\right)}$$

Cancel out the common shared variable n across both positions:

$$\lim_{n \rightarrow \infty} \frac{1}{2 + \frac{1}{n}}$$

Apply the limit condition boundaries as n approaches infinity:

$$\frac{1}{2 + \frac{1}{\infty}} = \frac{1}{2 + 0} = \frac{1}{2}$$

Compare the calculated limit value against the final checklist rule:

$$\frac{1}{2} < 1$$

Since the limit value satisfies $l < 1$, the series is **convergent**.